

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Ethical Hacking

COURSE CODE: ITA14

COURSE OUTCOME COVERED

CO2: Foot printing Tools: Conducting Competitive Intelligence, DNS Zone Transfers, Introduction to Social Engineering: Social Engineering Attacks and Countermeasures. (BT3)

Assessment Tool Weightage: 30%

Duration: 90 min

Total Marks: 50

Report Analysis Questions

1. A security analyst performed a network scan on the target 192.168.1.10 using the Nmap tool. The report identified several open ports, including HTTP (80), SSH (22), and MySQL (3306), along with the operating system and service versions. Analyze the report and explain the security risks associated with the identified services. Discuss how an attacker could exploit these services and recommend suitable security measures to protect the target system. **(20 Marks)**

Commands :

1. nmap scanme.nmap.org
2. nmap -sV scanme.nmap.org
3. sudo nmap -O scanme.nmap.org
4. sudo nmap -A scanme.nmap.org

Observation :

- The host is alive.
- HTTP (80) is open.
- SSH (22) is open.
- SIP ports 5060 and 5061 are open.
- Operating system information identified.

Security Recommendations :

- Keep all software updated.
- Enable firewall protection.

- Use strong authentication mechanisms.
- Perform periodic vulnerability assessments.
- Conduct regular penetration testing.

Report :

```

kali@kali ~
Session Actions Edit View Help
zsh: corrupt history file /home/kali/.zsh_history
kali@kali~$ nmap --version
Nmap version 7.95 ( https://nmap.org )
Platform: x86_64-pc-linux-gnu
Compiled with: liblua-5.4.7 openssl-3.5.4 libssh2-1.11.1 libz-1.3.1 libpcap-1.10.5 nmap-libidn2-2.3.4
Compiled without:
Available nsock engines: epoll poll select

kali@kali~$ nmap scanme.nmap.org
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-31 01:40 EDT
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.0022s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 987 filtered tcp ports (no-response)
PORT      STATE SERVICE
22/tcp    open  ssh
53/tcp    closed domain
80/tcp    open  http
139/tcp   closed netbios-ssn
143/tcp   closed imap
443/tcp   closed https
993/tcp   closed imaps
2022/tcp  closed down
3389/tcp  closed ms-wbt-server
5002/tcp  closed rfe
5060/tcp  open  sip
5061/tcp  open  sip-tls
8080/tcp  closed http-proxy

Nmap done: 1 IP address (1 host up) scanned in 88.31 seconds

kali@kali~$ nmap -sV scanme.nmap.org
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-31 01:43 EDT
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.0084s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 989 filtered tcp ports (no-response)
PORT      STATE SERVICE      VERSION
22/tcp    open  tcpwrapped
53/tcp    closed domain
80/tcp    open  http         Apache httpd 2.4.7
139/tcp   closed netbios-ssn
143/tcp   closed imap
443/tcp   closed https
3389/tcp  closed ms-wbt-server
5060/tcp  open  sip?
5061/tcp  open  sip-tls?
8080/tcp  closed http-proxy
8085/tcp  closed unknown

```

```

kali@kali ~
Session Actions Edit View Help
kali@kali~$ sudo nmap -O scanme.nmap.org
[sudo] password for kali:
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-31 01:49 EDT
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.011s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 985 filtered tcp ports (no-response)
PORT      STATE SERVICE      VERSION
22/tcp    open  ssh          OpenSSH 9.6p1 Ubuntu 0ubuntu0.23.04 (Ubuntu Linux; protocol 2.0)
53/tcp    closed domain
80/tcp    open  http         Apache httpd 2.4.7 ((Ubuntu))
139/tcp   closed mit-ml-dev
143/tcp   closed netbios-ssn
144/tcp   closed imap
443/tcp   closed https
500/tcp   closed isakmp
2022/tcp  closed down
3389/tcp  closed ms-wbt-server
5002/tcp  closed rfe
5060/tcp  open  sip
5061/tcp  open  sip-tls
8080/tcp  closed http-proxy
OS fingerprint not ideal because: Didn't receive UDP response. Please try again with -sSU
No OS matches for host

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 38.05 seconds

kali@kali~$ sudo nmap -A scanme.nmap.org
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-31 01:51 EDT
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.0027s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 991 filtered tcp ports (no-response)
PORT      STATE SERVICE      VERSION
80/tcp    open  http         Apache httpd 2.4.7 ((Ubuntu))
139/tcp   closed netbios-ssn
143/tcp   closed imap
443/tcp   closed https
993/tcp   closed imaps
3389/tcp  closed ms-wbt-server
5060/tcp  open  sip?
5061/tcp  open  sip-tls?
8080/tcp  closed http-proxy
Device type: bridge|VoIP adapter|general purpose
Running (JUST GUESSING): Oracle Virtualbox (94%), Slirp (94%), AT&T embedded (92%), QEMU (90%)
OS CPE: cpe:/o:oracle:virtualbox cpe:/a:danny_gasparovskii:slirp cpe:/a:qemu:qemu
Aggressive OS guesses: Oracle Virtualbox Slirp NAT bridge (94%), AT&T BGW210 voice gateway (92%), QEMU user mode network gateway (90%)
No exact OS matches for host (test conditions non-ideal).

```

```
kali@kali:~$ sudo nmap -A scanme.nmap.org -oN Assignment1_Nmap_Report.txt
Running (JUST GUESSING): Oracle Virtualbox (94%), Slirp (94%), AT&T embedded (92%), QEMU (90%)
OS CPE: cpe:/o:oracle:virtualbox cpe:/a:danny.gasparovski:slirp cpe:/a:qemu:qemu
Aggressive OS guesses: Oracle Virtualbox Slirp NAT bridge (94%), AT&T BGW210 voice gateway (92%), QEMU user mode network gateway (90%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop

TRACEROUTE (using port 80/tcp)
HOP RTT ADDRESS
1 1.77 ms scanme.nmap.org (45.33.32.156)

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 94.85 seconds

kali@kali:~$
kali@kali:~$ sudo nmap -A scanme.nmap.org -oN Assignment1_Nmap_Report.txt
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-31 01:54 EDT
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.0020s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 988 filtered tcp ports (no-response)
PORT      STATE SERVICE      VERSION
22/tcp    open  tcpwrapped
1.ssh-hostkey: ERROR: Script execution failed (use -d to debug)
53/tcp    closed domain
80/tcp    open  http         Apache httpd 2.4.7 ((Ubuntu))
1.http-server-header: Apache/2.4.7 (Ubuntu)
139/tcp   closed netbios-ssn
143/tcp   closed imap
161/tcp   closed snmp
443/tcp   closed https
2022/tcp  closed down
3389/tcp  closed ms-wbt-server
5860/tcp  open  sip
5861/tcp  open  sip-tls?
8080/tcp  closed http-proxy
Device type: bridge/VoIP adapter[general purpose]
Running (JUST GUESSING): Oracle Virtualbox (94%), Slirp (94%), AT&T embedded (92%), QEMU (90%)
OS CPE: cpe:/o:oracle:virtualbox cpe:/a:danny.gasparovski:slirp cpe:/a:qemu:qemu
Aggressive OS guesses: Oracle Virtualbox Slirp NAT bridge (94%), AT&T BGW210 voice gateway (92%), QEMU user mode network gateway (90%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop

TRACEROUTE (using port 80/tcp)
HOP RTT ADDRESS
1 0.40 ms scanme.nmap.org (45.33.32.156)

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 93.15 seconds

kali@kali:~$ ls
172.25.83.194 Assignment1_Nmap_Report.txt Desktop Documents Downloads ethical ethicalhacking index.html index.html.1 Music Pictures practical Public sample1.txt sample.txt Templates Videos
```

2. A phishing awareness campaign was conducted for employees of ABC Technologies Pvt. Ltd. using the GoPhish tool. The report indicates that 40% of employees clicked the phishing link, 20% entered their login credentials, and several users downloaded a malicious attachment. Analyze the report and identify the security weaknesses demonstrated by the employees. Recommend suitable user awareness programs, email security controls, and organizational policies to reduce the risk of successful social engineering attacks.(20 Mark).

Procedure :

1. Created Email Template.
2. Created Landing Page.
3. Configured Mailtrap SMTP.
4. Created User Group.
5. Created Campaign.
6. Send phishing emails.
7. Viewed campaign statistics.

Email Security Controls :

- Enable Spam Filtering.
- Use Email Gateways.
- Configure SPF (Sender Policy Framework).
- Configure DKIM (DomainKeys Identified Mail).
- Configure DMARC (Domain-based Message Authentication, Reporting, and Conformance).

- Block dangerous attachments.
- Scan email attachments using antivirus software.
- Enable URL filtering.

Report :

localhost

localhost

Access denied!

google classro...

CO2_AT1_Repor...

Releases - goph...

Gophish - Login

Users & Groups

Mailtrap - Email

+

-

□

×

https://mailtrap.io/sandboxes/projects

☆

☆

...

Chat

Venkat Karthik

Sandboxes

Projects

4

AI Assistant

Venkat Karthik

Email Sandboxes

Projects

Templates

Dev Tools

Integrations

API Tokens

Sandboxes usage

Emails sent0 / 50

Projects

My Project

Add Project

Add Sandbox

Sandboxes	Total Sent	Messages	Max size	Last message	Action
My Sandbox	0	0 / 0	10	Empty	

localhost x Access denied! x google classroo x CO2_AT1_Report x Releases - gophi x Gophish - Login x Dashboard - Go x Mailtrap - Email x +

Not secure https://127.0.0.1:3333

admin

gophish

- Dashboard
- Campaigns
- Users & Groups
- Email Templates
- Landing Pages
- Sending Profiles
- Account Settings
- User Management Admin
- Webhooks Admin
- User Guide
- API Documentation

Phishing Success Overview

% of Success

0 50 100

09:46:21.256

Email Sent Email Opened Clicked Link Submitted Data Email Reported

0 0 0 0 0

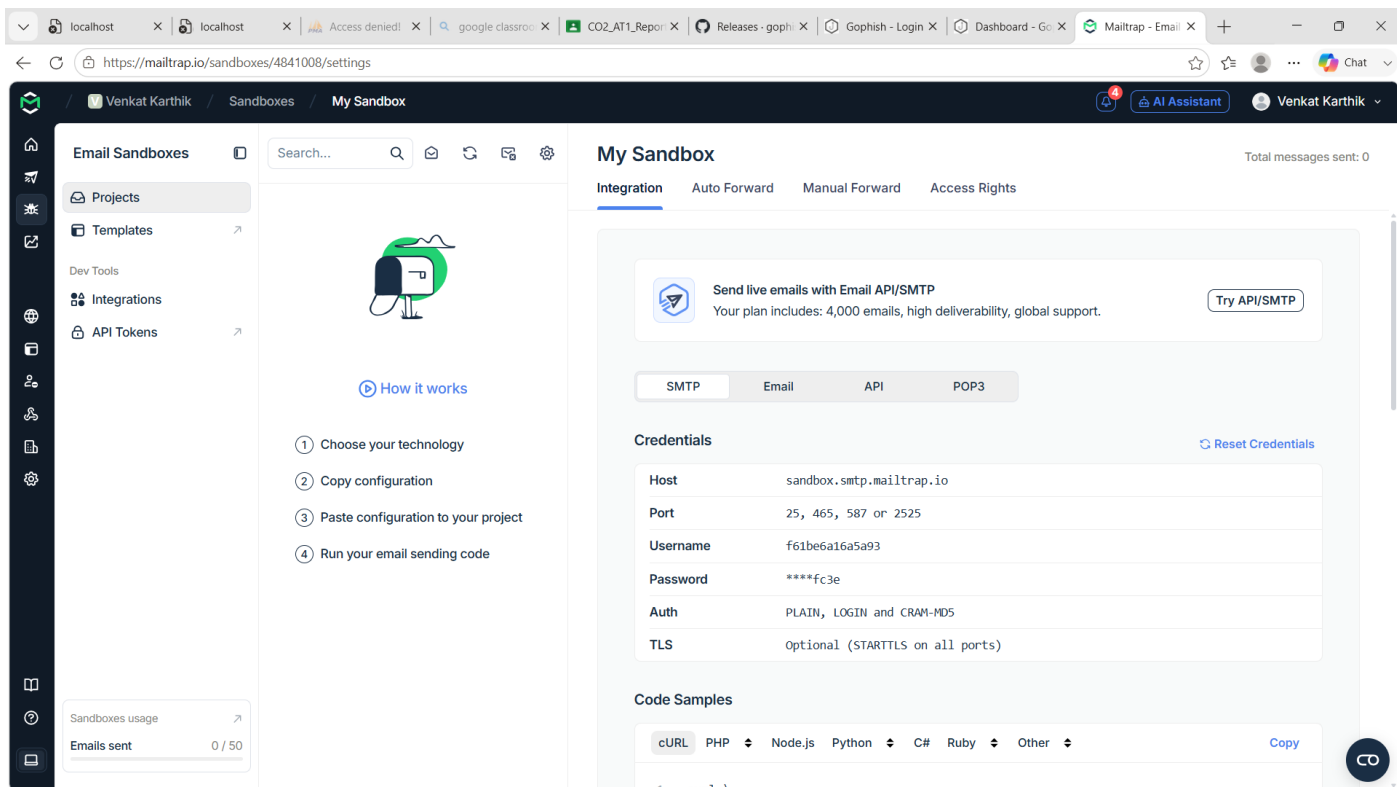
Recent Campaigns

View All

Show 10 entries

Search:

Name	Created Date						Status
ABC Technologies	August 6th 2026, 9:46:21 am	0	0	0	0	0	In progress



3. A DNS assessment report generated using the Dig tool shows that the organization's DNS server permits a successful DNS Zone Transfer (AXFR), exposing internal hostnames, mail servers, and DNS records. Analyze the report and explain the security implications of the exposed information. Discuss how attackers could exploit these findings and recommend best practices to secure the DNS infrastructure..(10 Marks)

Command :

dig AXFR zonetransfer.me @nsztml.digi.ninja

Observation :

- DNS Zone Transfer succeeded.
- Internal DNS records were exposed.
- Mail server records were visible.
- Multiple hostnames and IP addresses were disclosed.

Possible Attacks :

An attacker can use the exposed information to:

- Perform reconnaissance.

- Scan exposed servers.
- Launch phishing attacks.
- Target mail servers.
- Exploit vulnerable web servers.
- Attempt brute-force attacks on exposed services.
- Prepare for advanced cyberattacks.

To Secure DNS Infrastructure :

- Disable DNS Zone Transfer for unauthorized users.
- Allow AXFR only between trusted DNS servers.
- Configure Access Control Lists (ACLs).
- Enable DNSSEC to protect DNS integrity.
- Regularly update DNS software.
- Monitor DNS logs.
- Remove unnecessary DNS records.
- Perform regular DNS security audits.

```

kali@kali:~$ zsh; corrupt history file /home/kali/.zsh_history
(kali@kali)~$ dig AFR zonetransfer.me @nsztml.digi.ninja
; <>> DIG 9.20.15-2-Debian <>> AFR zonetransfer.me @nsztml.digi.ninja
;; global options: +cmd
zonetransfer.me. 7200 IN SOA nsztml.digi.ninja. robin.digi.ninja. 2019100801 172800 900 1209600 3600
zonetransfer.me. 7200 IN DNSKEY 256 3 7 AwAAAplo+IncBvXoI3d4Za+EDOfgnWQOIMFqJrImgZdsEeU8 ByHM0KsoxJOYV/7b3c8t1BmlQuqxZ7qH9CN7bFKD+bP/Bosn8b KgJyxEvElCj9+h/X0Y97IXaw9LmtJ6h4AxsrbgTr2g9KO1PSibvDPm WbQMaQk
zonetransfer.me. 7200 IN RRSIG 118K7R9ybOhwKIdf1yPKChokGcbYBPBDOTCNLD8oaZOWS/Rfl Bcz+zSYSGf6Lwlmdeulo7YAIII/AHQ7DsUsUzSzvGgRBsmaggytBN As8bTfZl5m+
zonetransfer.me. 301 IN TXT "google-site-verification:tp28J7AJUAH9fu23hMGccCOBI6XBmoVA04vMewxa"
zonetransfer.me. 7200 IN MX 0 ASPMX.L.GOOGLE.COM.
zonetransfer.me. 7200 IN MX 10 ALTI.ASPMX.L.GOOGLE.COM.
zonetransfer.me. 7200 IN MX 10 ALT2.ASPMX.L.GOOGLE.COM.
zonetransfer.me. 7200 IN MX 20 ASPMX2.GOOGLMAIL.COM.
zonetransfer.me. 7200 IN MX 20 ASPMX3.GOOGLMAIL.COM.
zonetransfer.me. 7200 IN MX 20 ASPMXA.GOOGLMAIL.COM.
zonetransfer.me. 7200 IN MX 20 ASPMXS.GOOGLMAIL.COM.
zonetransfer.me. 7200 IN A 5.196.105.14
zonetransfer.me. 7200 IN NS nsztml.digi.ninja.
zonetransfer.me. 7200 IN NS nsztml.digi.ninja.
zonetransfer.me. 7200 IN CERT PKIX 0 0 MIIIDTCACUQHFFH5BGzOrlyrxSh9lpimADUE29MAAGCSqGS1bl3QEeb CwUAMIGaMQswCYQDVQQGEWJHJEYMBGIAUECAwPU21dgggwYwN3o axJlMRIEAYDVQHDALTAgnvZellbGQxejAQBgNVBAQMCURpZ2luwSwQ YTEQM44
GAUUCwtsGsf2JuZezYMbGAIUEAwPeen9uZYKYw5Zwlv lML8MWdqYkoZThvCNAQRFGsg6dGIAZGIhaSSuaawQYTF4rwfyNYTA3 MOIAMZUIHTNAFTwbyNJTA3 MOIAMZUIHTNAMIGASQAQVQGVGGEJHQJYE MBYGIAUECAwPU21dgggwYwN3oaxJlMRIEAYDVQHDALTAgnvZell bGQxejAQBgNVBA
MEOIM0bMTU0MTUwMTU2fG4tY2F0YmFryMYEdQ 24kuK0RI8bPwQMSQMSWVLVB/FER566gvGqKW330TKATAVCH D9KTLfwfhv/fc4e68bmGvnyqJzfcm/SymamdgE8B67q35bY/vtUoh AGZ308ba6wBUH1j+341542KLvFEfc1PMQpfLgwPbjpcwC70 UBvQZFzbap+yv8f8Rlf6fV/
MaeIn9VS5hmUXtPd83PAwCtQB HJUvInBoAACwEAATAgBgABFAAOCAQKoxpfpv2/f tKxfCsMuab28DI2ULiAnbm+XYKSr6rborVatm0-xwD82ugvIQb uRLK9Lo5gm6VMWVI1INBKRSinsJe4tzewsNGrvfLvbbkULicEbt DElmk8rzEwU1MoSqt/dwg3GdcCM9Dp
KVhPh9uQ3pbys1qPL SApusJka31-Vruz3VweyETUKWAkn15v7Bg6F903W3JEIVZPBIC AdwfiYfx+tutEm612sxkl1pt/elb39cnKFqrs8BdzG3jyBC3CCYL mwANNU2ODNACOTuz8x621FsqVqJyfvyontPn4AcECYRWHe=
zonetransfer.me. 300 IN INFO "casio fr-RND "windows xp"
zonetransfer.me. 301 IN TXT "60a9b9b395xyvYPanQwUSSGavrbdlzeJeSZI"
sip.tcp.zonetransfer.me. 14000 IN SRV 0 0 5060 www.zonetransfer.me.
14.105.106.5.IN-ADDR.ARPA.zonetransfer.me. 7200 IN PTR www.zonetransfer.me.
asfbodathns.zonetransfer.me. 7900 IN AFSDB 1 asfbdbbox.zonetransfer.me.
asfbdbbox.zonetransfer.me. 7200 IN A 127.0.0.1
asfdbvolue.zonetransfer.me. 7800 IN AFSDB 1 asfbdbbox.zonetransfer.me.
canberra-office.zonetransfer.me. 7200 IN A 202.14.81.230
cmdexec.zonetransfer.me. 300 IN TXT "; ls"
contact.zonetransfer.me. 2592800 IN TXT "Remember to call or email Pippa on +44 125 4567890 or pippa@zonetransfer.me when making DNS changes"
deadbeef.zonetransfer.me. 7200 IN AAAA 123.128.181.132
deadbeef:: 1
dr.zonetransfer.me. 300 IN LOC 53 20 56.558 N 1 38 33.526 W 0.00m In 10000m 10m
DZC.zonetransfer.me. 7200 IN TXT "AbCdEfG"
email.zonetransfer.me. 2222 IN NAPTR 1 "p" "Edu-email" "" email.zonetransfer.me.zonetransfer.me.
email.zonetransfer.me. 7200 IN A 74.125.206.26
Hello.zonetransfer.me. 7200 IN TXT "Hi to Josh and all his class"
home.zonetransfer.me. 7200 IN A 127.0.0.1
Info.zonetransfer.me. 7200 IN TXT "ZoneTransfer.me service provided by Robin Wood - robin@digi.ninja. See http://digi.ninja/projects/zonetransferme.php for more information."
intnl.zonetransfer.me. NS
internal.zonetransfer.me. NS
intns1.zonetransfer.me. NS
intns1.zonetransfer.me. 300 IN A 82.1.108.41

```

```
kali@kali: ~$ dig axfr zonetransfer.me @nsztm1.digi.ninja > Assignment3_DNS_AXFR_Report.txt
;; Query time: 316 msec
;; SERVER: 81.4.108.41853(nsztm1.digi.ninja) (TCP)
;; WHEN: Fri Jul 31 06:34:39 EDT 2026
;; XFR size: 52 records (messages 1, bytes 3339)

zonetransfer.me. 300 IN HINFO "Casio fx-700G" "Windows XP"
_acme-challenge.zonetransfer.me. 301 IN TXT "60a0ShbUJ9xSvvy7pApQwCUsSgGxvrbDirjePEsZI"
sip.tcp.zonetransfer.me. 4000 IN SRV 0 0 5060 www.zonetransfer.me.
1a.1a5.19a.5.1a.400a.489a.zonetransfer.me. 7200 IN PTR www.zonetransfer.me.
asfdbauthdns.zonetransfer.me. 7900 IN AFSDB 1 asfdbbox.zonetransfer.me.
asfdbbox.zonetransfer.me. 7200 IN A 127.0.0.1
asfdrvolum.zonetransfer.me. 7200 IN AFSDB 2 asfdbbox.zonetransfer.me.
canberra-office.zonetransfer.me. 7200 IN A 202.14.81.210
cmdexec.zonetransfer.me. 300 IN TXT "i ls"
contact.zonetransfer.me. 2552000 IN TXT "Remember to call or email Pippa on +44 123 4507890 or pippa@zonetransfer.me when making DNS changes"
dc-office.zonetransfer.me. 7200 IN A 143.228.181.132
deadbeef.zonetransfer.me. 7201 IN AAAA dead:beaf::
dr.zonetransfer.me. 300 IN LOC 53 28 56.558 N 1 38 33.526 W 0.00m in 10000m 10m
DRC.zonetransfer.me. 7200 IN TXT "AmCdeGf"
email.zonetransfer.me. 2222 IN NAPTR 1 1 "p" "E2U+email" "" email.zonetransfer.me.zonetransfer.me.
email.zonetransfer.me. 7200 IN A 74.125.206.26
Hello.zonetransfer.me. 7200 IN TXT "Hi to Josh and all his class"
home.zonetransfer.me. 7200 IN A 127.0.0.1
Info.zonetransfer.me. 7200 IN TXT "ZoneTransfer.me service provided by Robin Wood - robin@digini.ninja. See http://digini.ninja/projects/zonetransferme.php for more information."
internal.zonetransfer.me. 300 IN NS intns1.zonetransfer.me.
internal.zonetransfer.me. 300 IN NS intns2.zonetransfer.me.
intns1.zonetransfer.me. 300 IN A 81.4.108.41
intns2.zonetransfer.me. 300 IN A 5.196.105.10
office.zonetransfer.me. 7200 IN A 4.23.39.254
ipvdactnow.org.zonetransfer.me. 7200 IN AAAA 2001:dfc1e8:11::c100:1332
owa.zonetransfer.me. 7200 IN A 207.46.197.32
robinwood.zonetransfer.me. 302 IN TXT "Robin Wood"
rp.zonetransfer.me. 321 IN RP robin.zonetransfer.me. robinwood.zonetransfer.me.
sip.zonetransfer.me. 3333 IN NAPTR 2 3 "p" "E2U+sip" "i".${sip:customer-service@zonetransfer.me}i .
sql.zonetransfer.me. 300 IN TXT "" or i i --"
sshock.zonetransfer.me. 7200 IN TXT "() { :|}; echo ShellShocked"
staging.zonetransfer.me. 7200 IN CNAME www.sydneyspeershouse.com.
alltcpportsopen.firewall.test.zonetransfer.me. 301 IN A 127.0.0.1
testing.zonetransfer.me. 301 IN CNAME www.zonetransfer.me.
vpn.zonetransfer.me. 4000 IN A 174.36.59.154
www.zonetransfer.me. 7200 IN A 5.196.105.10
xss.zonetransfer.me. 300 IN TXT "<script>alert('Boo')</script>"
zonetransfer.me. 7200 IN SOA nsztm1.digi.ninja. robin.digi.ninja. 2019100801 172800 900 1209600 3600
```

Code of Conduct:

I K. Venkata Karthik (192372092) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

K. Venkata Karthik
Signature of the student:

RUBRICS FOR CALCULATION

Report Analysis

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Report Analysis and Interpretation	25%	Accurately interprets the security report, identifies all key findings, and explains their significance with excellent technical understanding.	Correctly interprets most findings with minor omissions or inaccuracies.	Identifies only some findings and provides limited interpretation.	Unable to interpret the report or identify key findings.
Identification of Security Risks and Vulnerabilities	25%	Correctly identifies all security risks, vulnerabilities, and possible attack vectors with clear technical justification.	Identifies most risks and vulnerabilities with minor errors.	Identifies only a few risks with limited technical explanation.	Fails to identify major risks or vulnerabilities.
Application of Cybersecurity Concepts and Countermeasures	30%	Applies appropriate ethical hacking concepts and recommends comprehensive, practical, and technically sound countermeasures.	Recommends suitable countermeasures with minor technical gaps.	Provides limited or partially appropriate recommendations.	Recommendations are inappropriate, incomplete, or missing.
Technical Communication and Documentation	20%	Response is well-organized, technically accurate, clearly presented, and uses appropriate cybersecurity terminology throughout.	Response is organized and understandable with minor documentation or presentation issues.	Response lacks clarity, organization, or contains several technical inaccuracies.	Response is poorly organized, incomplete, and difficult to understand.